



COURSE SYLLABUS						
Cours e Code	ITEC 103	Course Title	Hardware/Software Installation and Maintenance		Course Credit	3 UNITS / 5 HOURS (2 Units Lecture And 1 Unit Lab)
Course Description		This course provides students with a practical and hands-on experience in installing, configuring, and maintaining computer hardware and software. Students will learn how to troubleshoot hardware and software problems, upgrade and repair computer systems, and implement security measures to protect computer systems from viruses, malware, and other security threats.				
Pre-Requisites		Basic Knowledge of computer systems and familiarity with microsoft windows operating system		Co-Requisites	None	
VISION		A Leading Comprehensive Polytechnic University in Asia				
MISSION		Advance an inclusive, equitable, and globally relevant polytechnic education towards national development.				
QUALITY POLICY STATEMENT		The Polytechnic University of the Philippines commits to provide inclusive and equitable quality education and promote lifelong learning opportunities for human advancement and social transformation through re-engineered polytechnic academic programs. Toward this end, we, the members of the PUP Community will vigorously and steadfastly endeavor to continuously improve the standard of University services to the satisfaction of our clients through the adoption and continuous review of our Quality Management System.				



INSTITUTIONAL LEARNING OUTCOMES (ILO)

As a polytechnic state university, PUP shall develop its students to possess:

1. **Critical and Creative Thinking.** Graduates use their rational and reflective thinking as well as innovative abilities to life situations in order to push boundaries, realize possibilities, and deepen their interdisciplinary, multidisciplinary, and/or transdisciplinary understanding of the world.
2. **Effective Communication.** Graduates apply the four macro skills in communication (reading, writing, listening, and speaking), through conventional and digital means, and are able to use these skills in solving problems, making decisions, and articulating thoughts when engaging with people in various circumstances.
3. **Strong Service Orientation.** Graduates exemplify strong commitment to service excellence for the people, the clientele, industry, and other sectors.
4. **Adept and Responsible Use or Development of Technology.** Graduates demonstrate optimized and responsible use of state- of-the-art technologies of their profession. They possess digital learning abilities, including technical, numerical, and/or technopreneurial skills.
5. **Passion for Lifelong Learning.** Graduates perform and function in society by taking responsibility in their quest for further improvement through lifelong learning.



	6. Leadership and Organizational Skills. Graduates assume leadership roles and become leading professionals in their respective disciplines by equipping them with appropriate organizational skills. 7. Personal and Professional Ethics. Graduates manifest integrity and adherence to moral and ethical principles in their personal and professional circumstances. 8. Resilience and Agility. Graduates demonstrate flexibility and the growth mindset to adapt and thrive in the volatile, uncertain, complex and ambiguous (VUCA) environment. 9. National and Global Responsiveness. Graduates exhibit a deep sense of nationalism as it complements the need to live as part of the global community where diversity is respected. They promote and fulfill various advocacies for human and social development.
COLLEGE GOALS AND OBJECTIVES	The Polytechnic University of the Philippines – Taguig Campus aims to achieve the following goals: 1. Offer curricula which are relevant and responsive to the changing needs of the industry and society; 2. Provide trainings to students that will enable them to become potent instruments to socio-economic development. 3. Prepare students to become productive, creative, innovative, and dynamic in their chosen fields of specialization. 4. Provide state-of-the-arts facilities of learning to optimize student development; 5. Inculcate appropriate values that are necessary to build a humane, disciplined, nationalist, and independent society; 6. Tap potentials of students, faculty, administrative staff and other stakeholders in formulating policies for institutional development; 7. Involve all stakeholders in enhancing programs for student’s faculty and employees’ development; 8. Partner with industry and other research institutions strengthening research capabilities of faculty, employees and students; 9. Facilitate presentation of research outputs in international fora, and their publication in recognized journals; 10.Undertake outreach and research-based extension programs by tapping the expertise and other resources of all stakeholders and, 11.Produce technologies for commercialization of livelihood improvement.



PROGRAM LEARNING OUTCOMES (PLO)	Based on CMO	Alignment to ILOs								
		1	2	3	4	5	6	7	8	9
	IT 01 Apply knowledge of computing, science and mathematics appropriate to the discipline									
	IT 03 Analyze complex problems, and identify and define the computing requirements appropriate to its solution									
	IT 04 Identify and analyze user needs to take them into account in the selection, creation, evaluation and administration of computer based systems									
	IT 05 Design, implement, and evaluate computer based systems, processes, components, or program to meet desired needs and requirements under various constraints									



	IT 06 Integrate IT-based solutions into the user environment effectively												
	IT 09 Assist in the creation of an effective IT project plan.												
	IT 10 Communicate effectively with the computing community and with society at large about complex computing activities through logical writing presentation and clear instruction												
	IT 08 Display high level of leadership and organizational skills in school and classroom management												
	IT 02 Understand best practices and standards and their applications												
	IT 07 Apply knowledge through the use of current techniques, skills, tools and practice necessary for the IT profession												
	IT 13 Recognize the need for and engage in planning self-learning and improving performance as a foundation for continuing professional development												
	IT 12 Understand professional, ethical, legal, security and social issues and responsibilities in the utilization of information technology												
	IT 11 Analyze the local and global impact of computing information technology on individual, organization and society												

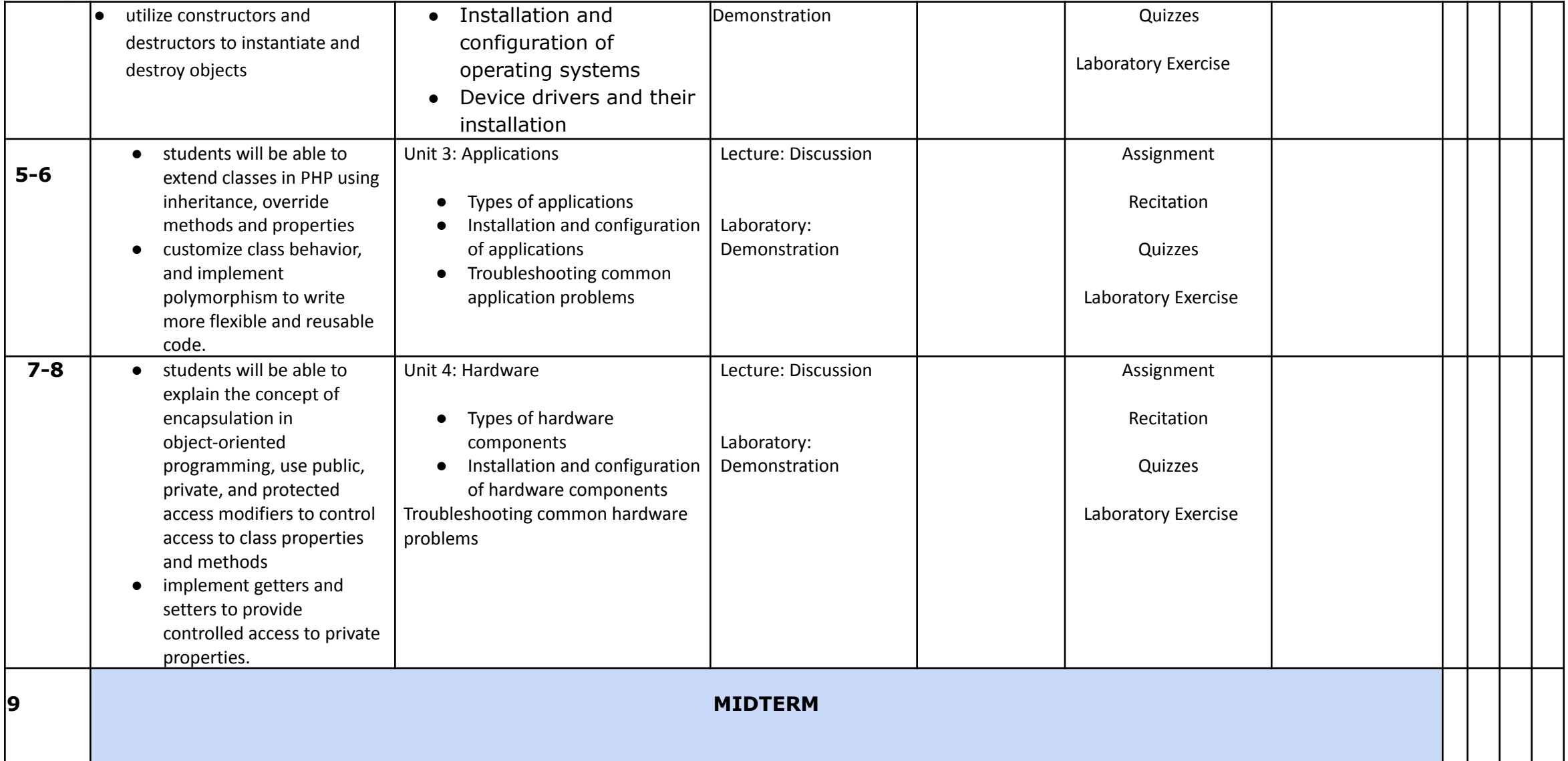


COURSE LEARNING OUTCOMES (CLO)		Alignment to PLOs							
		1	2	3	4	5	6	7	8
	1. Develop basic keying skills essential to effective production of personal and business letters, memoranda, manuscripts, and other reports.								
	2. Mastery of the different parts of the keyboard and their functions.								
	3. Reinforce Mastery of the alphabetic finger reaches, special characters, and symbols.								
	4. Format personal and business letters in block style and format simplified memoranda.								
	5. Produce mailable business letters using standard formats.								
	6. Display desirable personality traits and attitudes such as initiative, punctuality, critical judgement essential to efficient performance and success in keyboarding								
	7. Display desirable working habits like neatness, sense of responsibility to decide acceptable correspondence needed in the office.								
	8. Demonstrate good relations among students as they continue to develop the skills in keyboarding.								
	9. Employ keyboarding skills to advance in computer technology and software applications.								



OUTCOMES-BASED TEACHING AND LEARNING PLAN (OBTL PLAN)

Week	Desired Learning Outcomes (DLOs)	Learning Content/ Topics	Instructional Delivery Design		Assessment		Alignment to CLOs			
			Face-to-Face	Flexible Teaching and Learning Activities (FLTAs)	Face-to-Face	FLTAs	1	2	3	4
1-2	- Vision and Mission - Core Values - Content of the Syllabus - Grading System e. Be able to install the tools needed in this course f. students will be able to explain the concepts of classes, objects, and methods in object-oriented programming g. articulate the benefits of OOP over other programming paradigms h. write basic PHP code using object-oriented syntax.	Discuss the vision, mission, hymn and prayer of the school. Orient the class regarding the course syllabus, requirements, and the grading system. Unit 1: Introduction to Computer Systems - Components of a computer system - Functions of each component - Types of computer systems	Lecture: Discussion Laboratory: Demonstration		Assignment Recitation Quizzes Laboratory Exercise		X			
3-4	- students will be able to define classes in PHP - create objects from classes	Unit 2: Operating Systems Types of operating systems	Lecture: Discussion Laboratory:		Assignment Recitation					



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10-11	- students will be able to utilize advanced object-oriented programming concepts in PHP, including handling exceptions and errors, defining abstract classes and interfaces - implement design patterns such as the singleton, factory, and observer patterns.	Unit 5: Upgrades and Repairs - Hardware upgrades - Software upgrades - Repair of hardware and software components	Lecture: Discussion Laboratory: Demonstration		Assignment Recitation Quizzes Laboratory Exercise				
12-13	- students will be able to design and implement an application using OOP principles in PHP - demonstrate their understanding of key concepts and techniques learned throughout the course	Unit 6: Security - Types of security threats - Antivirus and anti-malware software - Firewalls and other security measures	Lecture: Discussion Laboratory: Demonstration						
14-15		Unit 7: Network Configuration and Maintenance - Types of networks - Network hardware components - Configuration of network settings - Troubleshooting common network problems - Implementing network security measures	Lecture: Discussion Laboratory: Demonstration						



16-17		Unit 8: Cloud Computing and Virtualization ● Introduction to cloud computing and virtualization ● Cloud computing models (SaaS, PaaS, IaaS) ● Virtualization technologies (Virtual Machines, Containers) ● Cloud deployment models (Public, Private, Hybrid) ● Cloud computing security and management	☐Lecture: Discussion Laboratory: Demonstration								
18	FINAL EXAMINATION										



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LIST OF REFERENCES

BOOKS:

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CLASSROOM POLICY	
FACE-TO-FACE DELIVERY	FLEXIBLE TEACHING AND LEARNING ACTIVITIES (FLTAs)
Please refer to the FLTA policies, except for the stipulation on Attendance. Attendance will be checked during face-to-face sessions.	Students’ Responsibilities Students are expected to: a. Take time to read and watch the provided class learning materials. b. Fulfill class requirements such as posting on discussion boards and submitting group presentations. c. Check the status of their submissions and inform the instructor if they are having problems submitting the requirements.



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The instructor may not be able to remind all students about their missing/unsubmitted requirements. If at the end of the course students still have missing requirements, the instructor will compute for the final grade without those submissions (They will be graded with 0) and submit the grade. If the final grade is "INC" grade, the student must contact the instructor to decide in completing the course, ASAP.

d. Confer with classmates or the instructor about the discussions during the synchronous sessions that they missed if unable to attend the live sessions. Recordings will be provided but may not be available immediately.

e. Inform the instructor of concerns that may prevent them from fully participating in the class. This includes withdrawal from the course, long term illnesses, personal matters, and others. Teachers will not be able to help you if you don't tell them of these concerns.

Attendance.

We recognize that students have varying access levels to technology and connectivity, which is why for the most part, the course is conducted asynchronously. During synchronous sessions, there will be no checking of attendance. However, everyone is encouraged to participate in these sessions. A recording and/or transcript of synchronous classes will be made available for students who are unable to attend a synchronous session.

Late Submissions.

Deadlines will be set for the submission of course requirements. This allows the instructor sufficient time to review them. It also helps students pace their work. If you experience problems submitting your work on time, please contact the instructor ahead of time. All requirements must be completed to pass the course. Late submissions will not merit deductions but may not receive timely feedback.

Cheating and Plagiarism.

Any work submitted by a student in this course for academic credit will be the student's own work. Any form of academic

dishonesty will be fully dealt with according to the stipulations of the PUP Student Handbook.

Participation.

Participation during discussions is greatly encouraged, and students should feel free to ask questions when an idea or lesson is unclear or confusing, especially during synchronous sessions.

Announcements.

Class announcements will be communicated through the class president. Other communication shall always be coursed through university-issued (i.e., official) email accounts. Students must use their PUP student accounts.

Intellectual Property and Copyright.

Original materials created for this online class is covered by university policies on intellectual property. Some (third-party) instructional materials (e.g., YouTube videos) might be under the public domain or are open access, while others (e.g., some assigned readings) might be subject to fair use. All third-party instructional materials will



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	be attributed to their original creators. Electronic copies of the assigned readings are for your private consumption only. Access to the course materials for this class for PUP students will be restricted to only those students registered in this course and will be available for only a
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2025
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WURI
The WORLD UNIVERSITY RANKINGS for INNOVATION

Rank 21
GENERATIVE AI

Rank 39
STUDENT SUPPORT

Rank 50
SYMBOL/



QS STARS

★★★★★
Employability
★★★★★
Teaching
★★★★★
Academic Development
★★★★★
Social Responsibility
★★★★★
Specialist Criteria: Bachelor of Science in Nutrition and Dietetics (BSND)
★★★★★
Inclusiveness
★★★★★
Internationalization



Times Higher Education Impact Rankings 2024
DEBUTS AT 1501+



	limited period. Please do not share these materials with anyone elsewhere. The same applies to the recordings of the synchronous sessions.	
COURSE REQUIREMENT/S	GRADING SYSTEM	
-	Mid Term Class Standing70% Quizzes (10 %) Attendance (5 %) Recitation (5 %) Activities/Assignments/Seat work/Projects (50 %) 30% Midterm Examinations 100%	Final Term Class Standing70% Quizzes (10 %) Attendance (5 %) Recitation (5 %) Activities/Assignments/Seatwork/Projects (50 %) 30% Final Examinations (Teaching Demonstration) 100%
	Final Grade = (Midterm Grade + Final Term Grade) / 2	
	CLASS INFORMATION	
	FACULTY INFORMATION	
	Course, Year and Section:	Name of Faculty:
Day and Time:	Consultation Time:	
Room:	Office Tel. No./ Mobile Phone No.	
School Year:	Semester:	E-Mail Address:



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★★★★★
Employability
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